Project Documentation & Executable Files Summary

Project: Human Development Index (HDI) Predictor using Machine Learning and Flask
This document combines the information requested in the Project Executable Files template and provides a concise project documentation suitable for SmartBridge submission.

1. Project Overview

The Human Development Index (HDI) Predictor is a Flask-based machine learning application that predicts the Human Development Index using socio-economic indicators. It includes data preprocessing, exploratory data analysis, model training, evaluation, and a web interface for making predictions.

2. Submission Checklist

Item	Status	Remarks
Complete Source Code	Yes	Included
README	Yes	Setup instructions available
requirements.txt	Yes	Python dependencies
Dataset	Yes	HDI dataset included
Environment Configuration	N/A	Not required
Flask Application	Yes	Implemented
Trained Model (.pkl)	Yes	Included
Test Results	Yes	Model evaluated
Documentation	Yes	Project reports included
Demo	Yes	Ready for presentation

3. Folder Structure

```
HDI/
├── Flask/
│   ├── app.py
│   ├── templates/
│   ├── static/
│   └── HDI.pkl
├── Training/
│   └── HumDevIndex.ipynb
├── documentation/
├── README.md
└── requirements.txt
```

4. Running the Project

- Clone or download the project.
- Create a virtual environment.
- Install dependencies using: pip install -r requirements.txt
- Run the application using: python app.py
- Open the local Flask URL in a browser.

- Select a country or provide input values to predict HDI.

5. Features

- Data preprocessing and cleaning
- Exploratory Data Analysis (EDA)
- Machine Learning model training
- HDI prediction
- Country-wise prediction
- Interactive Flask web application
- Visualization of dataset insights

6. Known Limitations

- Prediction quality depends on training data.
- Internet is not required for local execution.
- Future versions can include live datasets, authentication, and cloud deployment.